

CEMAC Jobs Analytics

Cameroon elasticities and WBES trade diagnostics

June 26, 2026

Roadmap

1. BDF Cleaning Results
2. Cameroun Census data on firm counts, employment, and revenue
3. Cameroun Sectoral Elasticities
4. B-READY Mapped WBES Constraints
5. CEMAC WBES Trade Analysis

Data cleaning overview

- The working panel is organized at the firm-year level, so duplicate handling changes both sample size and identification.
- Economic sector is based on the NACAM variable, which according to documentation its based on a Regional harmonized classifier linked to ISIC.
- The cleaning note isolates the two steps that most affect downstream interpretation: duplicate resolution and the NACAM-to-ISIC crosswalk.

Firm-Year Duplicate Cleaning

- Working key: `firmid × fin_yr`
- Perfect duplicates would be collapsed with original row order as the tie-breaker.
- Conflicting duplicate firm-years are fully removed from the cleaned analysis file, while we figure out the best approach for handling them.

	Value
Original observations	9,325
Perfect duplicate firm-years	0
Rows dropped from perfect duplicates	0
Conflicting duplicate firm-years	127
Rows dropped from conflicting duplicates	257
Final cleaned observations	9,068

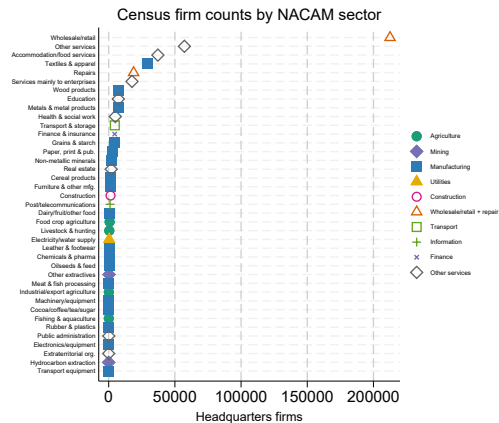
Cleaning Takeaways

- The cleaned panel resizes the sample from 9,325 to 9,068 observations after removing duplicate firm-years.
- There are 127 conflicting firm-years, accounting for 257 dropped obs. Those observations are excluded rather than resolved ad hoc.
- To avoid introducing unwanted noise we keep the analysis using NACAM classification and link it to ISIC aggregates when possible.

Cameroun Census Context

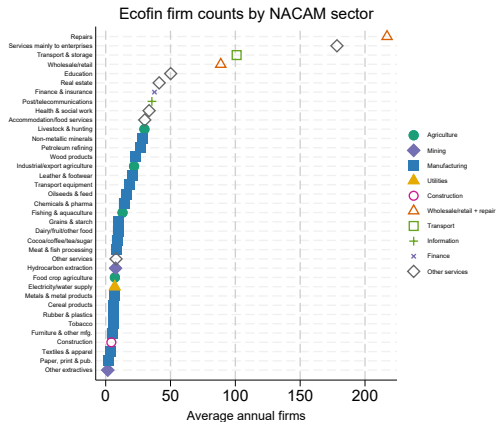
- The 2024 RGE census gives a whole-economy snapshot of establishments in Cameroon.
- Census activity labels are harmonized to the same NACAM sectors used in the administrative elasticity analysis.
- The next slides use headquarter-level data to show sector scale in firm counts, log employment, log annual revenue, and log revenue per worker.
- After this economy-wide view, the administrative tax panel focuses on formal firms observed over time, using the administrative data.

Census Firm Counts by Harmonized NACAM Sector



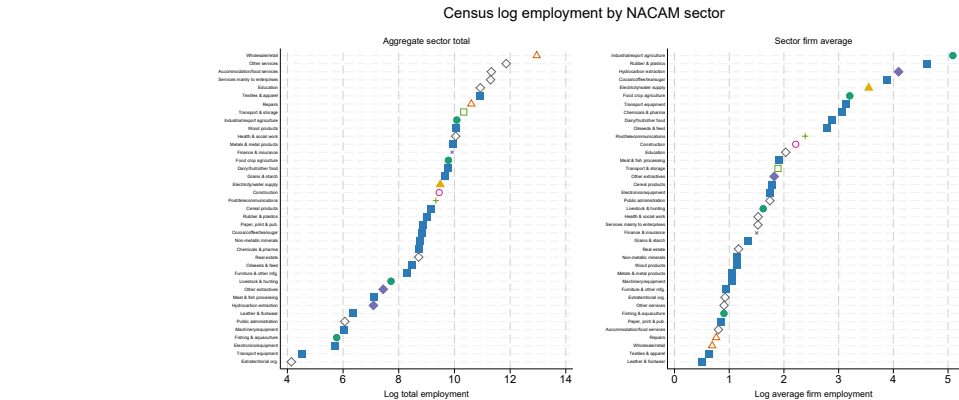
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Firm Counts by Harmonized NACAM Sector



Source: Cameroon administrative Ecofin panel; calculations by authors.

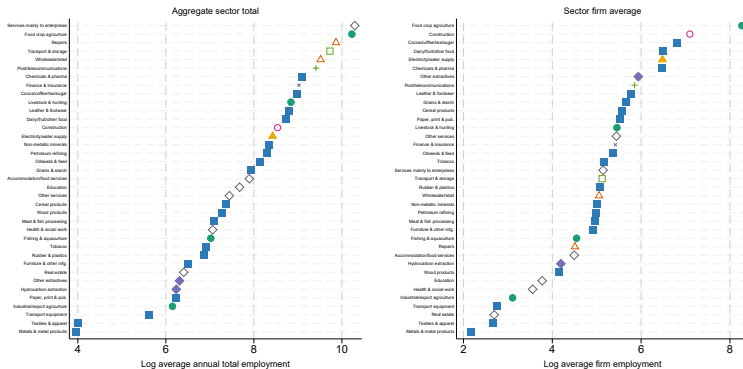
Census Log Employment by Harmonized NACAM Sector



Appendix: same figure in levels
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Log Employment by Harmonized NACAM Sector

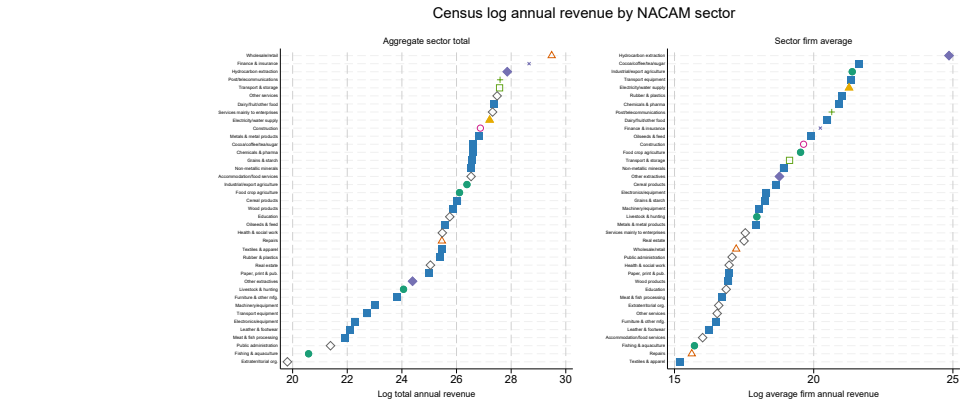
Ecofin log employment by NACAM sector



Appendix: same figure in levels

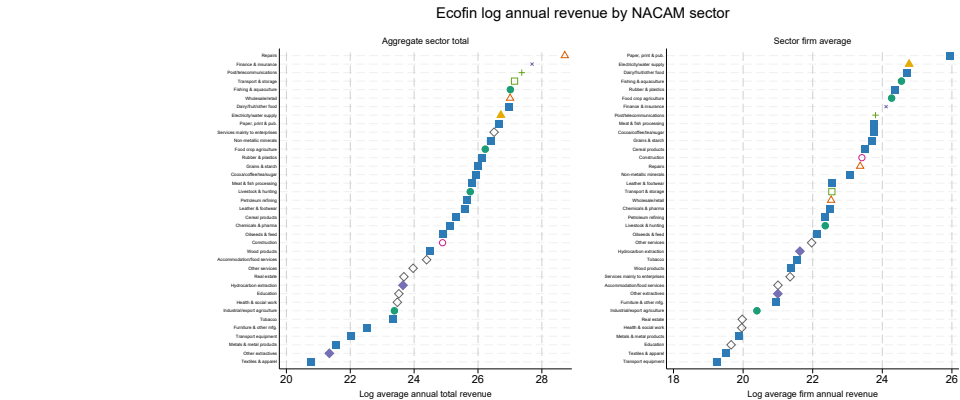
Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Log Revenue by Harmonized NACAM Sector



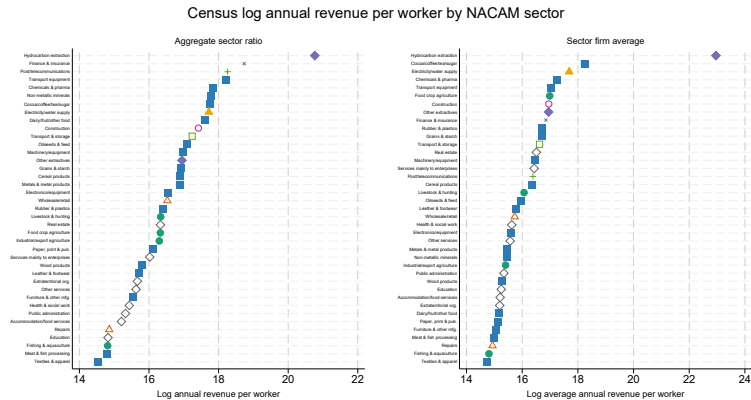
Appendix: same figure in levels
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Log Revenue by Harmonized NACAM Sector



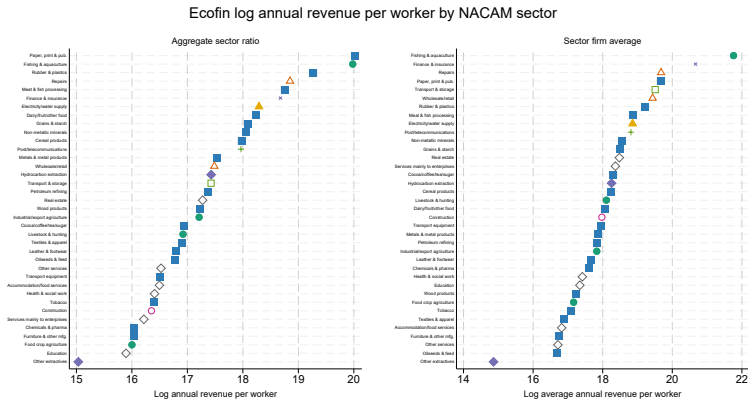
Appendix: same figure in levels
Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Log Revenue per Worker by Harmonized NACAM Sector



Appendix: same figure in levels
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Log Revenue per Worker by Harmonized NACAM Sector

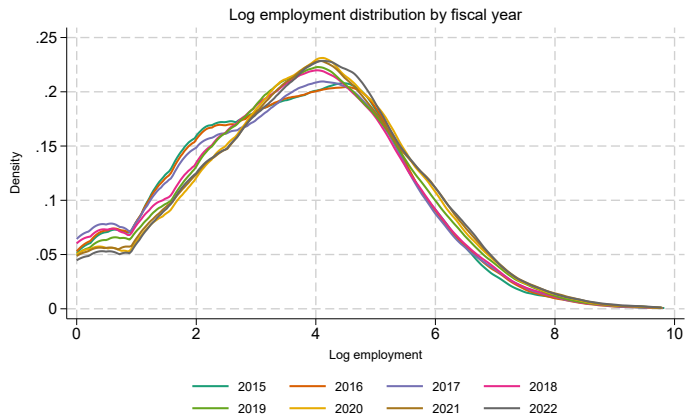


Specification and Sample

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s (\ln(X_{it}) \times D_{is}) + \varepsilon_{it} \quad (1)$$

- The unit of observation is a firm-year (i, t) .
- E_{it} is employment for firm i in year t .
- X_{it} is either value added or total revenue, estimated in separate regressions.
- D_{is} equals 1 if firm i belongs to NACAM sector s , and 0 otherwise.
- Sector is treated as a firm characteristic, so each firm contributes to one NACAM interaction term.
- α_i are firm fixed effects; δ_{st} are NACAM-sector-by-fiscal-year fixed effects.
- β_s is the elasticity for firms in sector s , identified through the interaction between $\ln(X_{it})$ and the sector indicator D_{is} .
- No additional firm-level controls are included beyond these fixed effects.
- Standard errors are clustered at the firm level; reported sectors have at least 30 observations and 10 firms in both outcome samples.

Employment Distribution by Fiscal Year



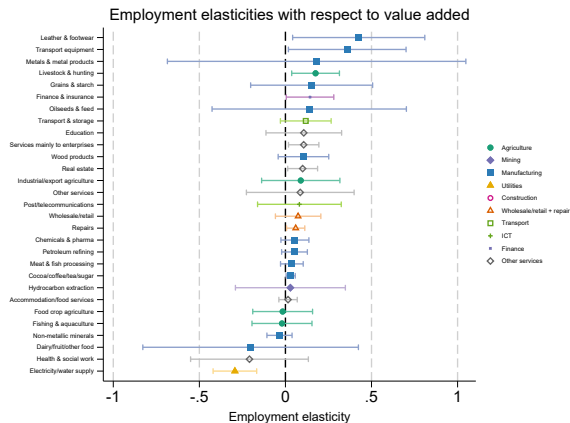
Source: Cameroon administrative Ecofin panel; calculations by authors.

Value-Added Elasticity Specification

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s^{VA} (\ln(VA_{it}) \times D_{is}) + \varepsilon_{it} \quad (2)$$

- Estimates use the Cameroon administrative firm-year panel.
- Firm fixed effects absorb time-invariant firm differences at the taxpayer/firm level.
- NACAM-sector-by-fiscal-year fixed effects absorb annual shocks common to firms in the same detailed activity.
- No age, ownership, exporter-status, or other firm controls are included.
- The plotted point for each sector is β_s^{VA} with a firm-clustered 95 percent interval.

Value-Added Elasticities Across NACAM Sectors



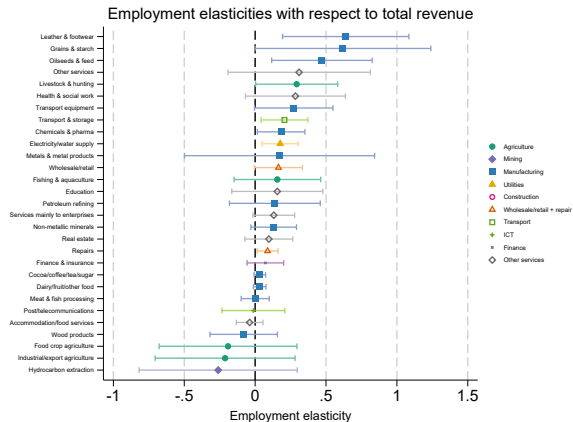
Source: Cameroon administrative Ecofin panel; calculations by authors.

Total-Revenue Elasticity Specification

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s^R (\ln(R_{it}) \times D_{is}) + \varepsilon_{it} \quad (3)$$

- R_{it} is total revenue, estimated in a separate regression from value added.
- Fixed effects are at the firm level and NACAM-sector-by-fiscal-year level.
- No additional firm-level controls are included beyond these fixed effects.
- The plotted point for each sector is β_s^R with standard errors clustered at the firm level.

Total-Revenue Elasticities Across NACAM Sectors

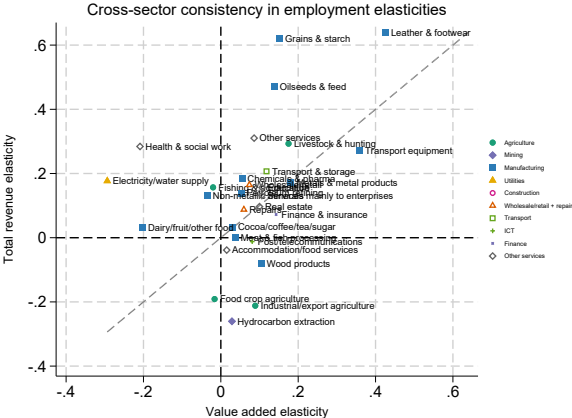


Source: Cameroon administrative Ecofin panel; calculations by authors.

Cross-Model Comparison

- The next plot compares the sector-specific value-added and total-revenue elasticities.
- Each point is one NACAM sector retained by the common inclusion rule.
- Points near the 45-degree line indicate sectors where both revenue proxies give similar employment responses.
- Large gaps flag sectors where value added and gross revenue tell different stories.

Cross-Model Consistency

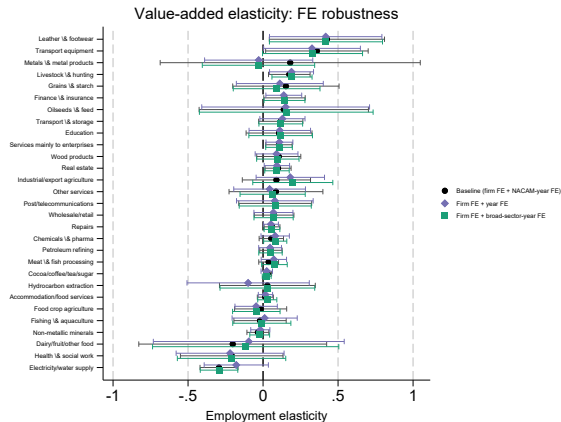


Source: Cameroon administrative Ecofin panel; calculations by authors.

Fixed-Effect Robustness Check

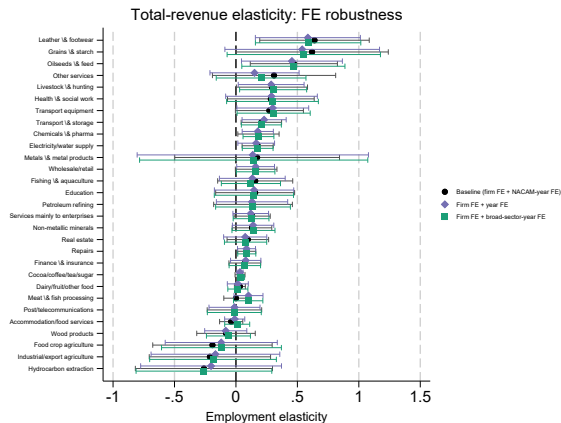
- FE means fixed effects. All robustness specifications keep taxpayer/firm fixed effects and firm-clustered standard errors.
- Baseline: firm FE plus detailed NACAM-sector-by-fiscal-year FE.
- Looser alternative: firm FE plus fiscal-year FE only.
- Intermediate alternative: firm FE plus broad-activity-group-by-fiscal-year FE.
- No age, ownership, exporter-status, or other firm-level controls are added in these robustness checks.

Value-Added Elasticity Robustness



Source: Cameroon administrative Ecofin panel; calculations by authors.

Total-Revenue Elasticity Robustness



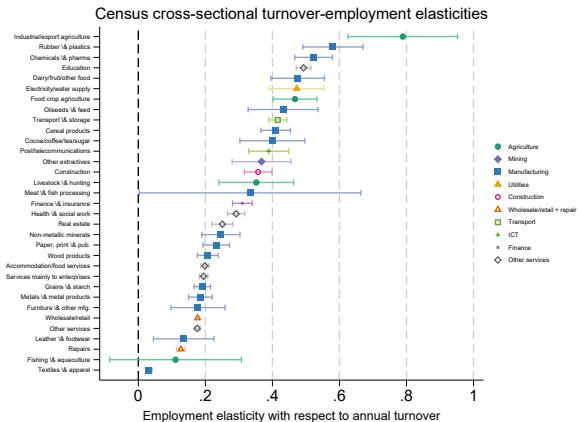
Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Turnover-Employment Specification

$$\ln(E_i) = \alpha_s + \sum_s \beta_s [\ln(R_i) \times D_{is}] + \varepsilon_i \quad (4)$$

- The unit of observation is a Census headquarters firm i .
- E_i is employment and R_i is annual turnover from RGE3.
- D_{is} equals 1 if firm i belongs to NACAM sector s .
- α_s are detailed NACAM sector intercepts.
- β_s is the cross-sectional employment elasticity with respect to annual turnover in sector s .
- There are no firm fixed effects, year fixed effects, or NACAM-year fixed effects because the Census has no panel time variation.
- Standard errors are heteroskedasticity-robust; reported sectors have at least 30 positive-employment and positive-turnover headquarters firms.

Census Turnover-Employment Elasticities



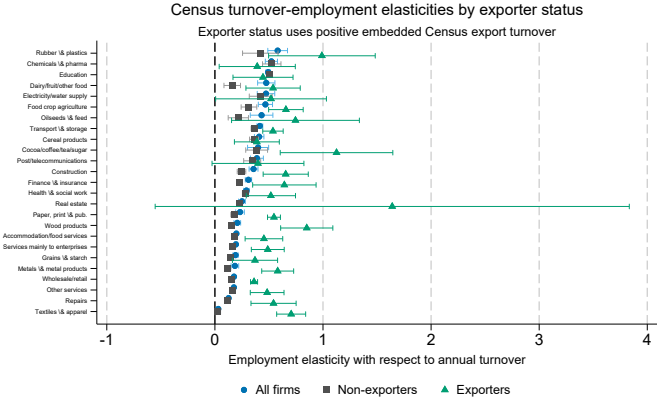
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Exporter-Interaction Specification

$$\ln(E_i) = \alpha_s + \beta_s \ln(R_i) + \gamma_s \text{Exporter}_i + \delta_s [\ln(R_i) \times \text{Exporter}_i] + \varepsilon_i, \quad i \in s \quad (5)$$

- The unit of observation is a Census headquarters firm i in NACAM sector s .
- E_i is employment and R_i is annual turnover from RGE3.
- Exporter_i equals 1 when embedded Census export turnover is positive.
- β_s is the non-exporter elasticity; $\beta_s + \delta_s$ is the exporter elasticity.
- The total elasticity is estimated separately among all firms in the same reported sector sample.
- Standard errors are heteroskedasticity-robust.
- Reported sectors have at least 30 usable firms, 10 exporters, and 10 non-exporters.

Census Turnover Elasticities by Exporter Status

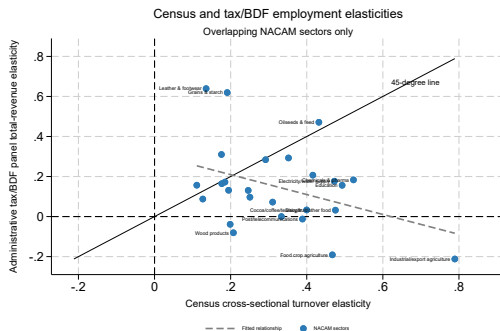


Cross-sectional Census headquarters firms. Robust 95 percent confidence intervals.

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Turnover-Employment Robustness

- Census estimates are cross-sectional slopes; Ecofin estimates are panel total-revenue elasticities with firm and NACAM-year fixed effects.
- The solid black line marks equality between the two estimates; the dashed grey line shows the fitted relationship across overlapping NACAM sectors.



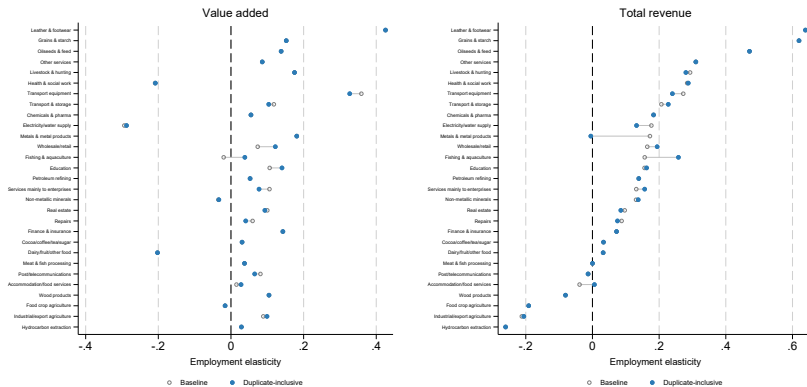
Sources: INS Cameroon, RGE3 (2023-2024), and Cameroon administrative Ecofin panel; calculations by authors.

Duplicate Robustness Check

- The baseline specification excludes conflicting duplicate firm-years from the cleaned panel.
- The robustness specification keeps those duplicate rows as separate observations.
- Firm fixed effects and clustered standard errors still use the original firm identifier, so duplicate rows do not become new firms.
- The next plot compares baseline and duplicate-inclusive elasticities sector by sector.

Baseline vs Duplicate-Inclusive Elasticities

Baseline vs duplicate-inclusive employment elasticities



Source: Cameroon administrative Ecofin panel; calculations by authors.

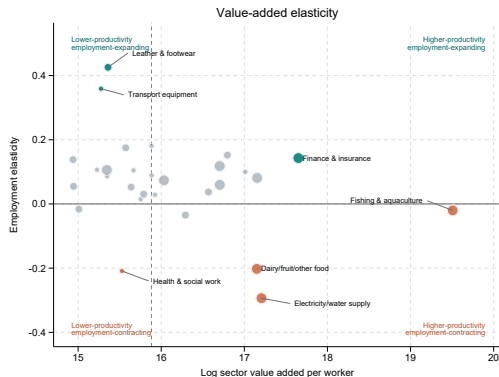
Elasticity, Productivity, and Sector Scale

- The opportunity maps compare sector employment elasticities with log sector value added per worker.
- The solid horizontal line separates employment-expanding and employment-contracting sectors; the dashed vertical line marks median productivity.
- Bubble size identifies the lower, middle, or upper tercile of each scale measure across the 29 plotted sectors; legends report the numerical ranges.
- Labels identify the two highest and two lowest elasticities in each panel plus the highest-productivity sector.

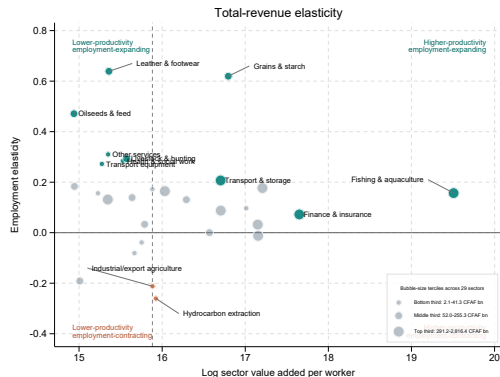
Opportunity Map: Total Sector Revenue

Where are Cameroon's employment-expanding sectors?

Employment response, productivity, and sector scale



Vertical line marks median log sector value added per worker.

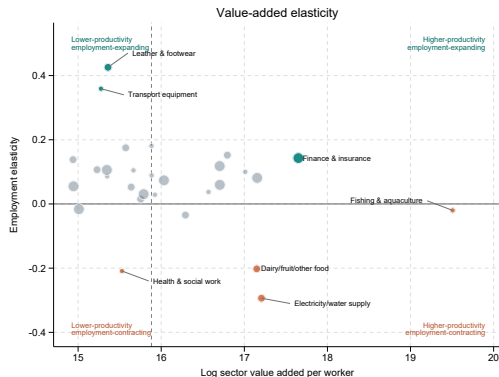


Source: Cameroon administrative Ecofin panel; calculations by authors.

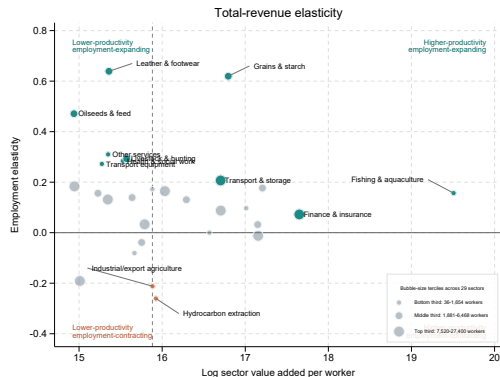
Opportunity Map: Total Sector Employment

Where are Cameroon's employment-expanding sectors?

Employment response, productivity, and sector scale



Vertical line marks median log sector value added per worker.

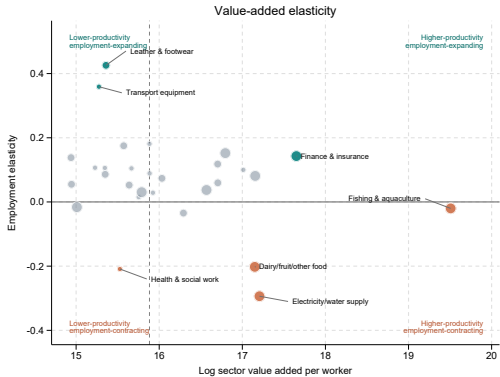


Source: Cameroon administrative Ecofin panel; calculations by authors.

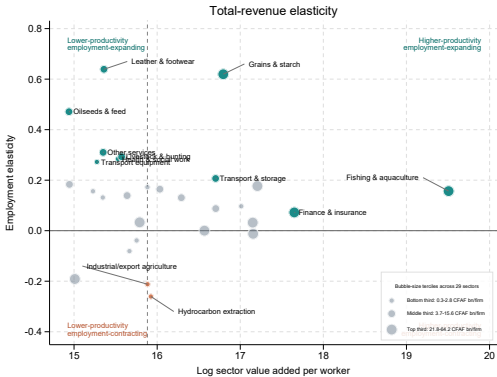
Opportunity Map: Average Firm Revenue

Where are Cameroon's employment-expanding sectors?

Employment response, productivity, and sector scale



Vertical line marks median log sector value added per worker.

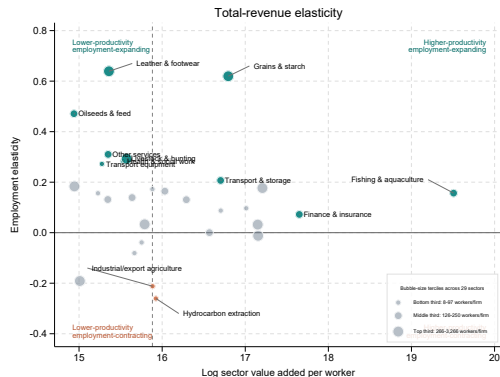
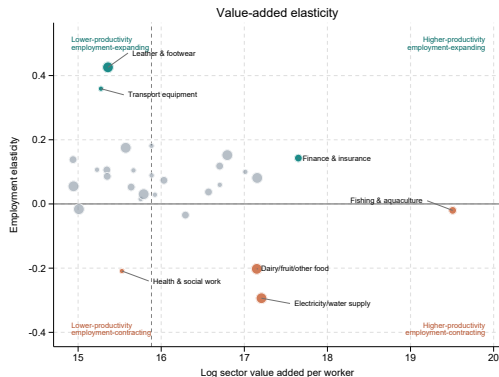


Source: Cameroon administrative Ecofin panel; calculations by authors.

Opportunity Map: Average Firm Employment

Where are Cameroon's employment-expanding sectors?

Employment response, productivity, and sector scale



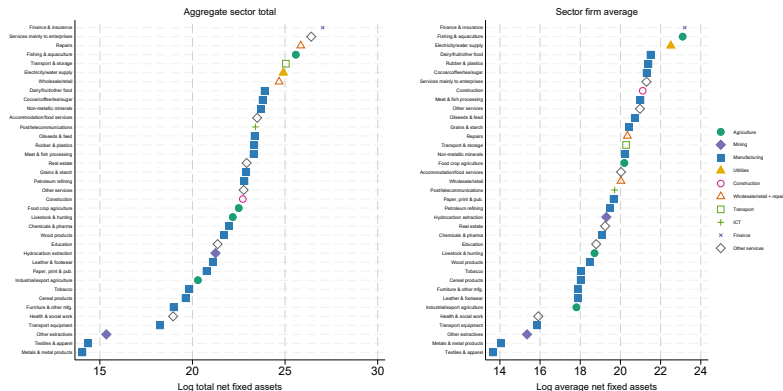
Source: Cameroon administrative Ecofin panel; calculations by authors.

Administrative Ecofin Asset Diagnostics

- The RGE/Census file does not include the dictionary-listed asset field, so asset diagnostics use the administrative Ecofin panel instead.
- Net fixed assets are the preferred capital proxy because they most closely represent firm capital stock in the current BDF variables.
- Each firm contributes once to the sector totals, using its latest available positive asset observation.
- Net total assets are shown as a broader balance-sheet comparison.

BDF Net Fixed Assets by Harmonized NACAM Sector

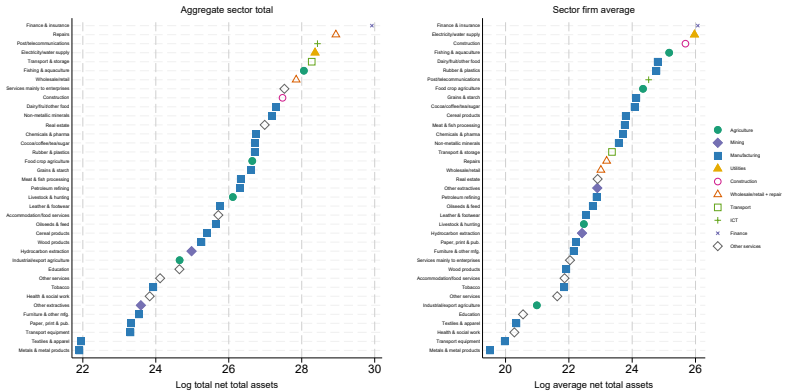
BDF log net fixed assets by NACAM sector



Source: Cameroon administrative Ecofin panel; calculations by authors.

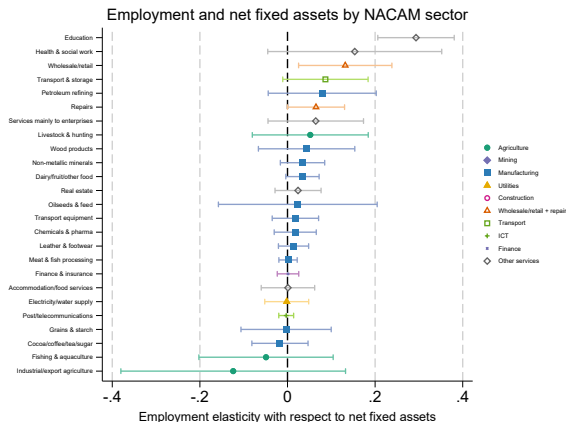
BDF Net Total Assets by Harmonized NACAM Sector

BDF log net total assets by NACAM sector



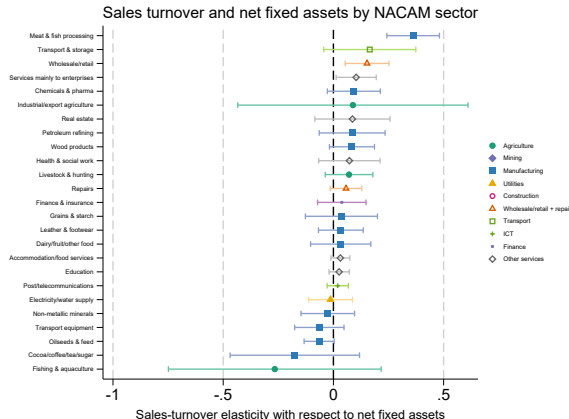
Source: Cameroon administrative Ecofin panel; calculations by authors.

Employment Response to Net Fixed Assets



Source: Cameroon administrative Ecofin panel; calculations by authors.

Sales-Turnover Response to Net Fixed Assets



Source: Cameroon administrative Ecofin panel; calculations by authors.

Revenue-Elasticity Sector Deciles

	Revenue decile	Total revenue elasticity	Value added elasticity	Revenue observations	VA observations
Leather & footwear	10	0.639	0.426	164	148
Grains & starch	10	0.620	0.152	78	67
Oilseeds & feed	9	0.471	0.138	128	115
Other services	9	0.310	0.086	60	55
Livestock & hunting	9	0.293	0.175	235	197
Health & social work	8	0.284	-0.209	263	243
Transport equipment	8	0.272	0.359	145	125
Transport & storage	8	0.207	0.118	788	708
Chemicals & pharma	7	0.183	0.055	110	100
Electricity/water supply	7	0.177	-0.294	56	54
Metals & metal products	7	0.172	0.181	53	45
Wholesale/retail	6	0.164	0.074	654	575
Fishing & aquaculture	6	0.157	-0.020	79	71
Education	6	0.156	0.107	368	364
Petroleum refining	5	0.139	0.053	215	196
Services mainly to enterprises	5	0.131	0.106	1,387	1,264
Non-metallic minerals	5	0.131	-0.035	222	183
Real estate	4	0.097	0.100	325	301
Repairs	4	0.088	0.060	1,712	1,522
Finance & insurance	4	0.072	0.143	300	272
Cocoa/coffee/tea/sugar	3	0.033	0.031	70	57
Dairy/fruit/other food	3	0.032	-0.202	76	64
Meat & fish processing	3	0.000	0.037	65	43
Post/telecommunications	2	-0.012	0.081	281	258
Accommodation/food services	2	-0.039	0.015	241	224
Wood products	2	-0.081	0.105	183	171
Food crop agriculture	1	-0.191	-0.016	54	51
Industrial/export agriculture	1	-0.212	0.089	174	134
Hydrocarbon extraction	1	-0.260	0.029	59	48

Decile 10 is highest; decile 1 is lowest.

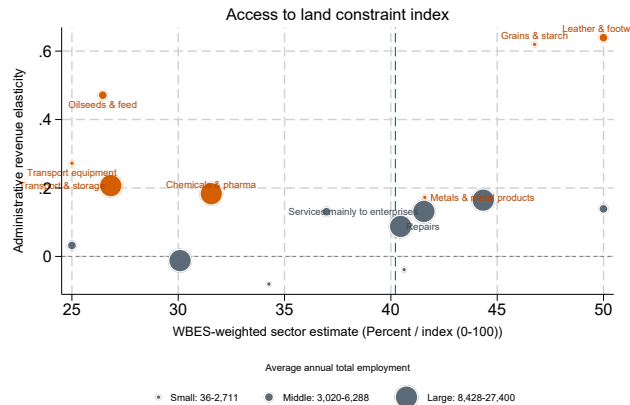
Main Takeaways

- Elasticities vary meaningfully across sectors rather than lining up around a single Cameroon-wide coefficient.
- Negative or weak value-added elasticities remain concentrated in a small set of sectors, which is useful for prioritizing follow-up diagnostics.

B-READY/WBES Constraint Extension

- The analysis uses the 16 questions marked `Priority = yes` in the reviewed B-READY workbook.
- Cameroon WBES firms are mapped from four-digit ISIC Rev.4 main-product codes to reviewed NACAM sectors using the official INS bridge.
- The horizontal axis shows WBES-weighted sector constraints; the vertical axis shows administrative total-revenue employment elasticities.
- Bubble area shows average annual total sector employment using fixed groups across indicators, not WBES precision.
- Orange sectors exceed the published Cameroon constraint benchmark and have an employment elasticity of at least 0.20.
- Labels identify opportunity sectors and the two largest-employment sectors with valid indicator responses.
- Results are descriptive, not causal; sectors without indicator responses are omitted from the corresponding chart.
- Fourteen indicators reproduce the national B-READY value within two points. The selected market-concentration item is unavailable in Cameroon microdata, and the VAT-refund denominator requires further review.

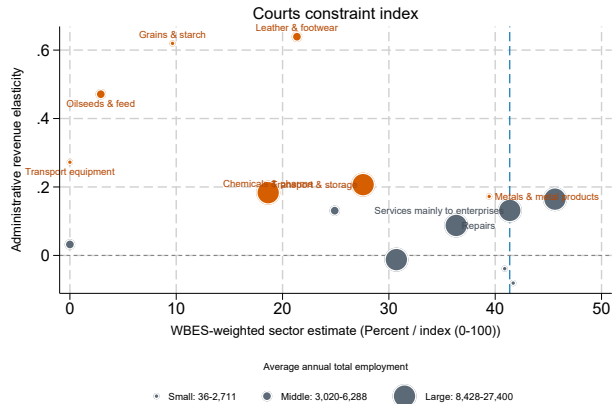
Access to Land Constraints by Sector



Dashed vertical line: published Cameroon B-READY benchmark. Orange: revenue elasticity declines 7-10. Descriptive; sectors without indicator responses omitted.

Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

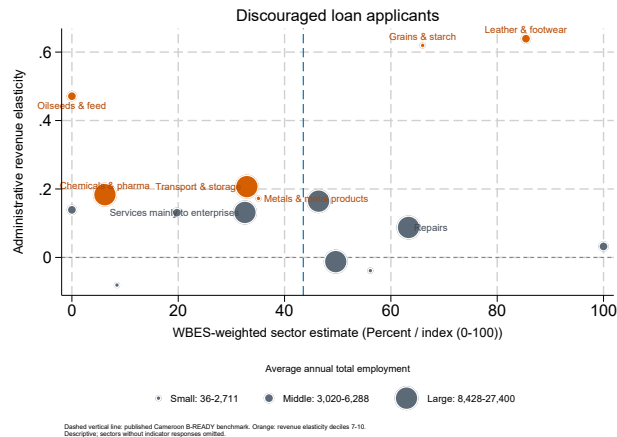
Courts Constraints by Sector



Dashed vertical line: published Cameroon B-READY benchmark. Orange: revenue elasticity declines 7-10. Descriptive; sectors without indicator responses omitted.

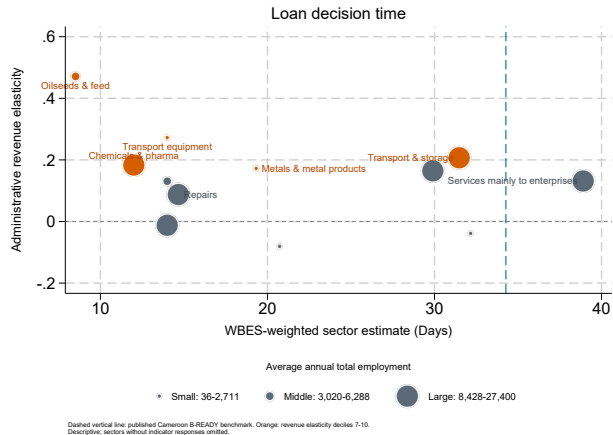
Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Discouraged Loan Applicants by Sector



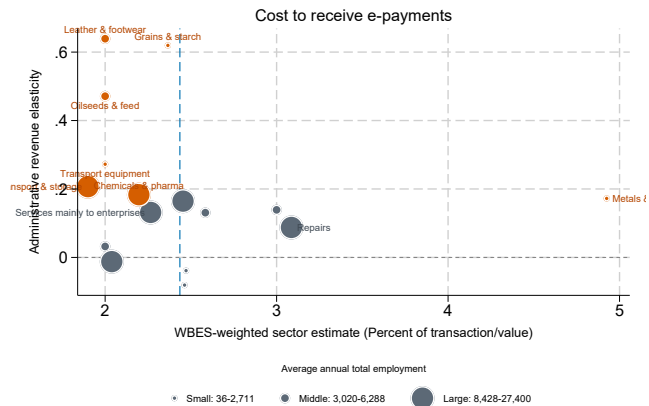
Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Loan Decision Time by Sector



Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

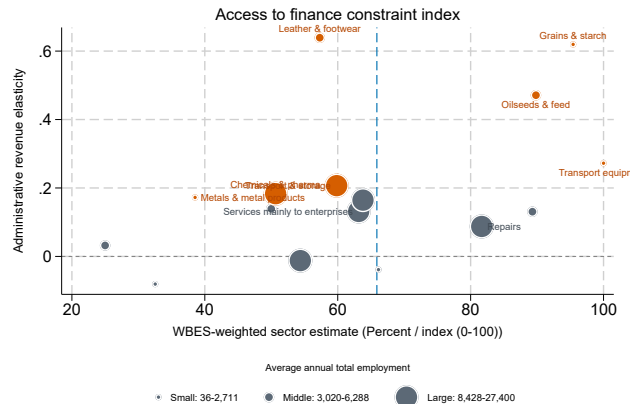
Cost to Receive E-Payments by Sector



Dashed vertical line: published Cameroon B-READY benchmark. Orange: revenue elasticity declines 7-10. Descriptive; sectors without indicator responses omitted.

Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

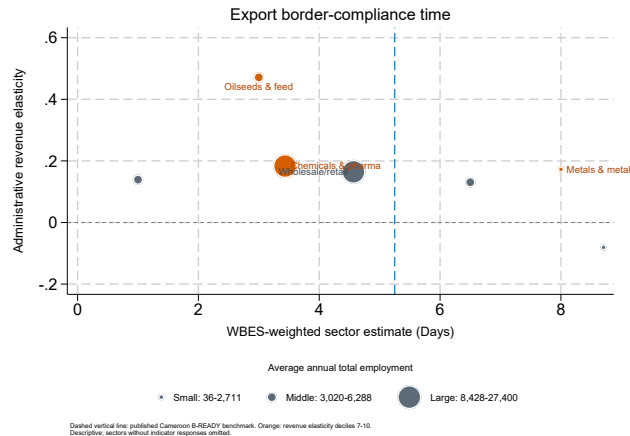
Access to Finance Constraints by Sector



Dashed vertical line: published Cameroon B-READY benchmark. Orange: revenue elasticity deciles 7-10. Descriptive; sectors without indicator responses omitted.

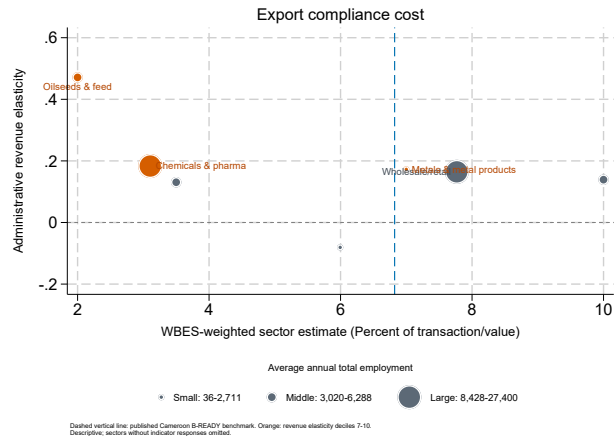
Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Export Border-Compliance Time by Sector



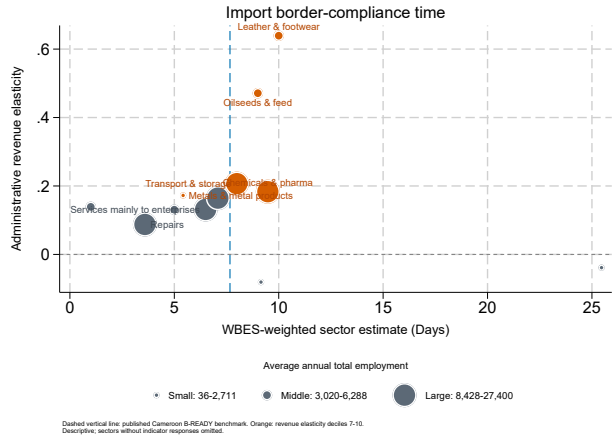
Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Export Compliance Cost by Sector



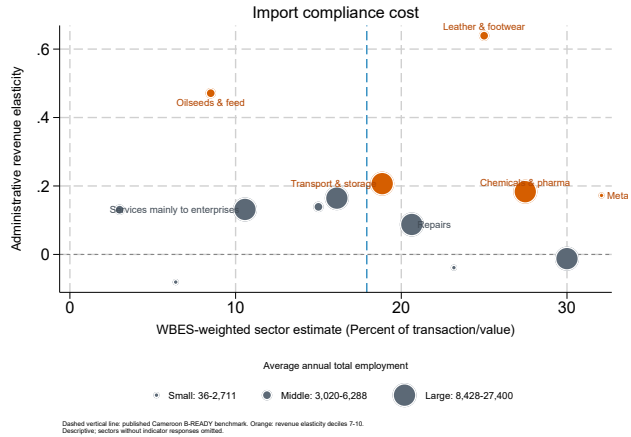
Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Import Border-Compliance Time by Sector



Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Import Compliance Cost by Sector



Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

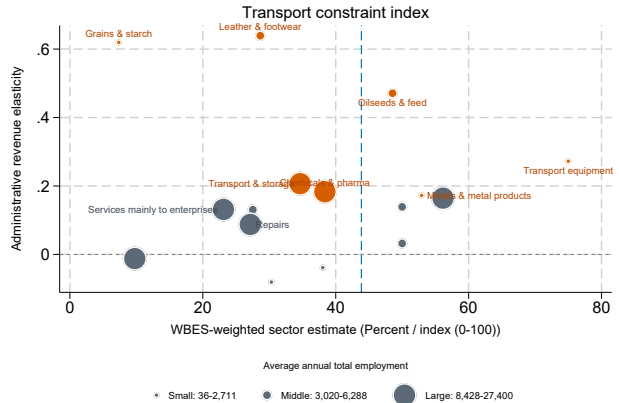
Customs and Trade Constraints by Sector



Dashed vertical line: published Cameroon B-READY benchmark. Orange: revenue elasticity declines 7-10. Descriptive; sectors without indicator responses omitted.

Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

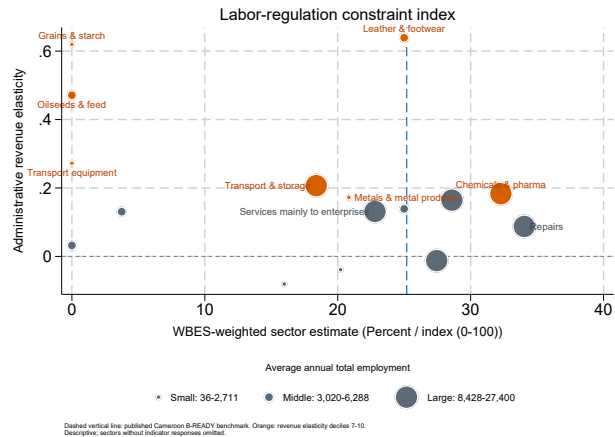
Transport Constraints by Sector



Dashed vertical line: published Cameroon B-READY benchmark. Orange: revenue elasticity declines 7-10. Descriptive; sectors without indicator responses omitted.

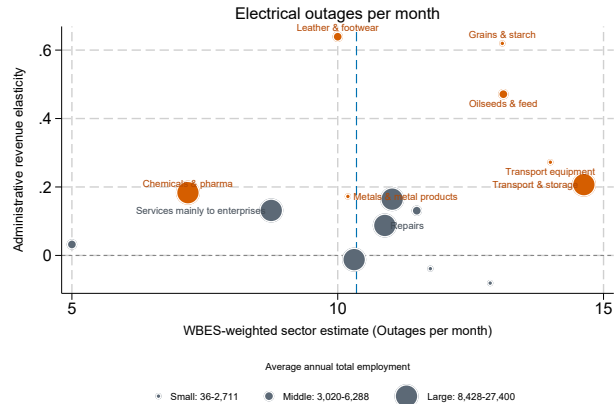
Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Labor-Regulation Constraints by Sector



Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Electrical Outages by Sector



Dashed vertical line: published Cameroon B-READY benchmark. Orange: revenue elasticity declines 7-10. Descriptive; sectors without indicator responses omitted.

Sources: World Bank Enterprise Surveys, Cameroon 2024, and Cameroon administrative Ecofin panel; calculations by authors.

Phase 3: WBES Trade Diagnostics

- Phase 3 uses the latest available World Bank Enterprise Survey wave for CEMAC countries.
- The first step is a cleaning and availability audit for sales, employment, sector, export, import, and input-origin variables.
- Gabon is not retained in the main descriptive panels because its latest available WBES wave is 2009 and has no ISIC Rev.4 coverage.
- The WBES formal-sector sampling frame covers non-agricultural private firms, so sector rows use eligible activity aggregates rather than agriculture.

WBES Descriptive Statistics

	Cameroon	Central Afr. Rep.	Chad	Congo	Eq. Guinea	SSA excl. CEMAC	High-income countries
Sample composition							
Fiscal year	2023	2022	2022	2023	2023	2022	2023
Surveyed firms	615	151	164	365	173	18252	30855
Firms represented (000s)	5	5	1	2	1	354	6879
	14.6	28.9	5.3	12.9	15.8	11.9	18.8
% Exporting firms	[2.1]	[6.9]	[2.5]	[2.8]	[3.6]	[0.6]	[0.6]
	17.7	15.2	28.9	29.8	43.3	24.6	16.3
% Manufacturing	[2.7]	[5.1]	[4.4]	[3.6]	[4.5]	[0.8]	[0.5]
	8.8	15.0	3.1	4.0	7.8	10.0	15.1
% Construction/utilities	[1.7]	[5.5]	[2.3]	[1.3]	[2.7]	[0.7]	[0.6]
	61.7	67.3	59.5	61.4	34.0	60.9	53.3
% Trade/hospitality/transport	[3.3]	[6.8]	[5.3]	[3.9]	[4.6]	[1.0]	[0.8]
	11.9	2.5	8.5	4.8	14.8	4.5	15.4
% Other services	[2.0]	[1.0]	[3.5]	[1.9]	[3.2]	[0.4]	[0.7]
Economic activity (% total sales)							
	17.2	15.6	16.7	11.8	27.6	31.6	24.4
Manufacturing	[3.2]	[5.1]	[5.7]	[3.9]	[6.6]	[2.6]	[1.2]
	10.5	23.0	3.5	5.9	25.8	16.1	14.1
Construction/utilities	[3.3]	[11.6]	[2.4]	[2.8]	[9.4]	[2.4]	[1.4]
	59.7	45.9	70.1	78.6	35.1	48.8	45.7
Trade/hospitality/transport	[5.5]	[10.0]	[7.8]	[6.2]	[8.9]	[2.7]	[1.8]
	12.6	15.4	9.7	3.7	11.5	3.6	15.8
Other services	[4.4]	[10.8]	[4.4]	[3.4]	[6.2]	[0.7]	[1.5]
Average sales (current USD thousands)							
	447.3	36.0	312.5	1058.7	1075.0	763.3	6325.6
All firms	[44.3]	[6.9]	[53.1]	[226.4]	[184.5]	[38.3]	[209.6]
	795.4	47.1	894.1	3588.8	1002.5	1771.2	12779.7
Exporting firms	[166.3]	[19.1]	[444.0]	[1439.2]	[522.5]	[164.6]	[631.7]
	387.4	31.9	280.2	726.7	1088.0	633.6	4801.6
Non-exporting firms	[43.6]	[7.3]	[41.5]	[164.6]	[197.2]	[36.8]	[208.5]
Average employment							
	27.8	22.3	15.2	40.9	32.0	44.9	39.2
All firms	[2.8]	[5.3]	[1.6]	[9.0]	[4.4]	[4.9]	[4.3]
	51.3	14.7	23.0	151.5	19.4	93.5	73.8
Exporting firms	[7.3]	[2.9]	[6.6]	[57.0]	[7.4]	[19.5]	[13.8]
	23.8	24.5	14.7	26.4	34.4	38.9	31.1
Non-exporting firms	[3.1]	[8.3]	[1.7]	[4.7]	[4.9]	[5.1]	[4.2]

Sales use fiscal-year official exchange rates. Brackets report standard errors.

WBES Country Coverage

	Firms	SurveyYear	FiscalYear	FirmsWithISIC4	Retained
Cameroon	615	2,024	2,023	615	1
Central African Republic	151	2,023	2,022	151	1
Chad	164	2,023	2,022	164	1
Congo	365	2,024	2,023	365	1
Equatorial Guinea	173	2,024	2,023	173	1
Gabon	179	2,009	.	0	0

Retained excludes Gabon from the main descriptive panels because its latest available wave is 2009 and lacks ISIC Rev.4.

Core Variable Availability

	Firms	Sales	SalesW	Employment	Weight	ISIC4	ExportShare	ImportStatus	ImportValue	FirmAge	OwnershipControls	SalesShareOff100	InputShareOff100	OwnershipShareOff100
Cameroon	615	614	614	615	615	615	615	613	275	614	614	0	0	0
Central African Republic	151	139	139	151	151	151	134	126	56	147	128	0	0	0
Chad	164	164	164	164	164	164	164	164	110	164	164	0	0	0
Congo	365	352	352	358	365	365	353	331	123	363	360	0	0	0
Equatorial Guinea	173	172	172	173	173	173	173	173	108	173	173	0	0	0

Counts are unweighted establishments in retained CEMAC latest-wave surveys.

Negative Special-Code Audit

	Employment	PermanentEmp	Sales	DomesticShare	IndirectExport	DirectExport	DomesticInput	ForeignInput	DirectImport	RawMatCost	ResaleCost
Cameroon	0	0	1	0	0	0	2	2	1	12	1
Central African Republic	0	0	12	11	16	14	25	25	19	3	0
Chad	0	0	0	0	0	0	0	0	0	0	0
Congo	0	7	13	6	10	10	34	34	1	27	42
Equatorial Guinea	0	0	1	0	0	0	0	0	0	0	0

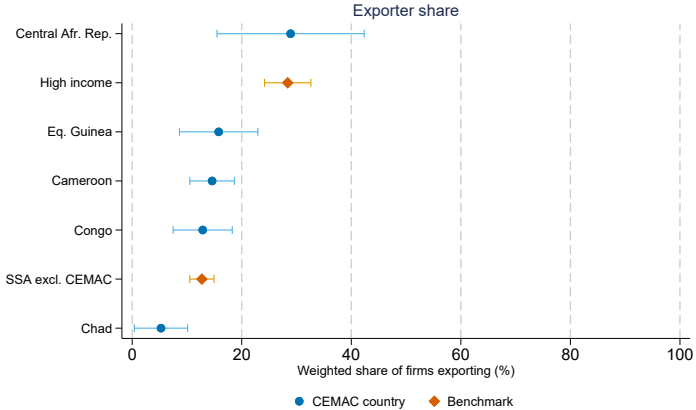
Negative WBES special codes are audited before being recoded to missing for nonnegative levels, percentages, and binary responses.

Winsorization Cutoffs

	SalesP5	SalesP95	InputCostP5	InputCostP95
Cameroon	9,125,300	2930040000	2,000,000	2800000000
Central African Republic	1,744,000	435,000,000	300,000	50,000,000
Chad	6,750,000	1115225000	1,380,000	500,000,000
Congo	1,000,000	12000000000	150,000	600,000,000
Equatorial Guinea	13,000,000	9090000000	300,000	669,000,000

Sales and input costs are winsorized at the 5th and 95th percentiles within each country wave.

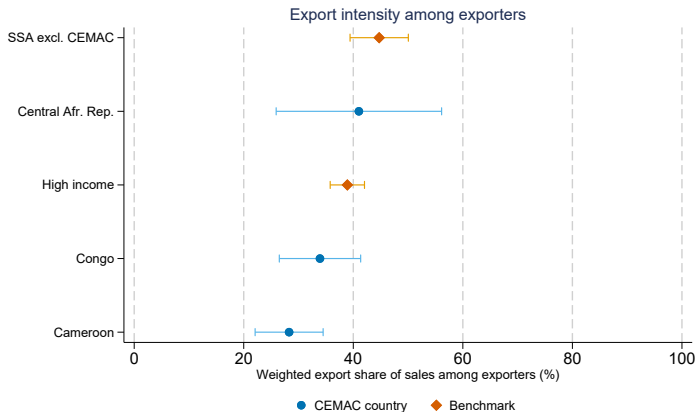
Exporter Participation



WBES weights used. Benchmark points are equal-country averages. Gabon is coverage-only; rows with N<30 suppressed.

Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

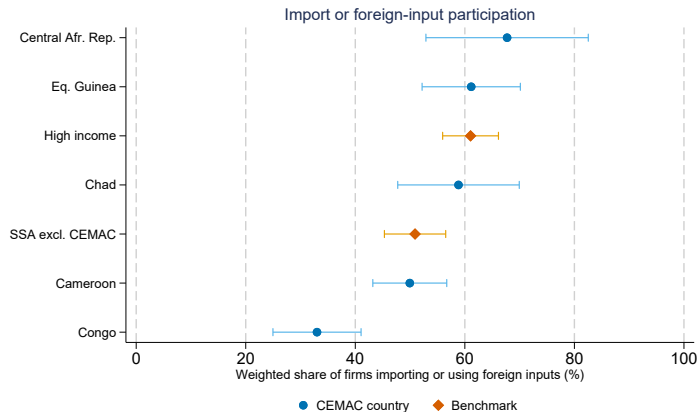
Export Intensity Among Exporters



WBES weights used. Benchmark points are equal-country averages. Gabon is coverage-only; rows with N<30 suppressed.

Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Import and Foreign-Input Participation



WBES weights used. Benchmark points are equal-country averages. Gabon is coverage-only; rows with N<30 suppressed.

Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Two-Way Trader Status



WBES weights used. Benchmark points are equal-country averages. Gabon is coverage-only; rows with N<30 suppressed.

Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Focal WBES Revenue-Exporter Specification

$$\ln(E_i) = \alpha_c + \beta_c \ln(R_i) + \gamma_c \text{Exporter}_i + \delta_c [\ln(R_i) \times \text{Exporter}_i] + \theta_c Z_i + \lambda_{cq(i)} + \varepsilon_i, \quad i \in c \quad (6)$$

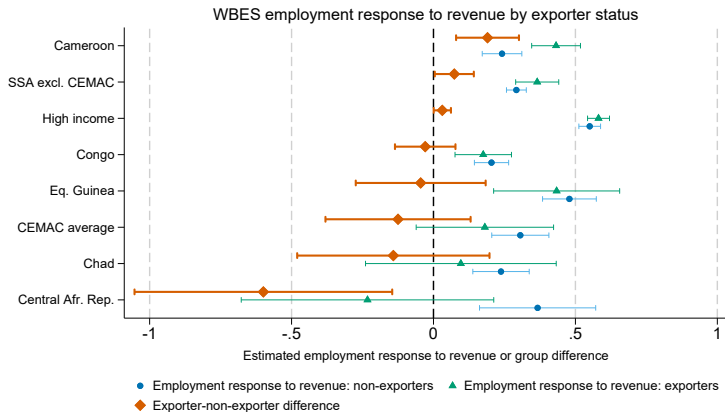
- c indexes the country. The model is estimated separately within each country.
- WBES is cross-sectional, so there are no firm fixed effects or panel controls.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{cq(i)}$ are country-specific ISIC Rev.4 section fixed effects.
- Estimates use WBES weights and robust standard errors, not clustered standard errors.
- β_c is the non-exporter slope; $\beta_c + \delta_c$ is the exporter slope.
- The focal estimate is δ_c , the exporter minus non-exporter slope difference.
- Display rule: at least 30 firms overall and at least 10 firms in each exporter group.

Country Revenue-Exporter Interaction Estimates

	Non-exporter	Exporter	Difference	SE	Lower 95	Upper 95	P-value	N	Exporters	Non-exporters
Cameroon	0.241	0.432	0.190	0.056	0.080	0.301	0.001	612	136	476
Central Afr. Rep.	0.367	-0.232	-0.599	0.229	-1.053	-0.145	0.010	108	36	72
Chad	0.238	0.096	-0.141	0.171	-0.480	0.197	0.411	164	13	151
Congo	0.204	0.175	-0.029	0.054	-0.136	0.077	0.591	334	46	288
Eq. Guinea	0.479	0.434	-0.045	0.116	-0.274	0.184	0.698	172	18	154
CEMAC average	0.306	0.181	-0.125	0.130	-0.381	0.131	0.338	1,390	249	1,141
High income	0.550	0.581	0.031	0.015	0.001	0.061	0.045	28,692	10,697	17,995
SSA excl. CEMAC	0.292	0.365	0.073	0.035	0.005	0.142	0.036	14,990	2,693	12,297

The difference column is the focal exporter minus non-exporter slope estimate.

Country Employment Responses to Revenue by Exporter Status



Weighted WBES cross-section. Controls: firm age, ownership shares, and ISIC section FE.

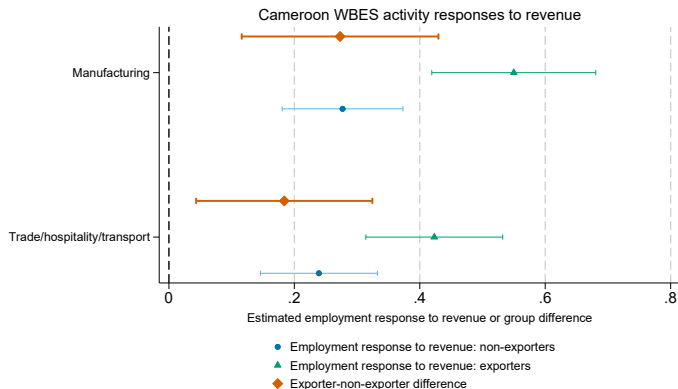
Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Cameroon WBES Activity-Group Specification

$$\ln(E_i) = \alpha_a + \beta_a \ln(R_i) + \gamma_a \text{Exporter}_i + \delta_a [\ln(R_i) \times \text{Exporter}_i] + \theta_a Z_i + \lambda_{aq(i)} + \varepsilon_i, \quad i \in a \quad (7)$$

- a indexes broad Cameroon WBES activity groups; the model is estimated separately within each group.
- Groups combine ISIC Rev.4 sections: manufacturing; construction/utilities; trade/hospitality/transport; and other services.
- WBES is cross-sectional, so there are no firm fixed effects.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{aq(i)}$ are activity-specific ISIC Rev.4 section fixed effects where there is within-activity variation.
- Estimates use WBES weights and robust standard errors, not clustered standard errors.
- Only two groups satisfy the display rule. The excluded groups are construction/utilities with 8 exporters and other services with 9, below the required 10 exporters.

Cameroon Activity Employment Responses to Revenue (Diagnostic)



Weighted WBES cross-section. Controls: firm age, ownership shares, and ISIC section FE.

Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Activity estimates are exploratory diagnostics; suppressed groups remain documented in the generated sample audit.

Supporting WBES Export-Value Specification

$$\ln(E_i) = \alpha + \beta \ln(X_i^+) + \gamma \text{Exporter}_i + \theta Z_i + \lambda_{q(i)} + \varepsilon_i \quad (8)$$

- These supporting models use export value rather than total revenue.
- Latest-wave WBES estimates remain weighted cross-sectional associations.
- X_i^+ is export value, with log export value set to zero for non-exporters in the all-firm model.
- The exporter dummy separates the extensive margin from the positive-export-value slope.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{q(i)}$ are ISIC Rev.4 section fixed effects.
- Rows with fewer than 30 observations are suppressed; benchmark rows are equal-country averages.

WBES Country Trade-Employment Estimates

	All-firm slope	SE	N	Exporters	Exporter-only slope	SE	N	Countries
Cameroon	0.325	0.038	613	136	0.373	0.040	136	1
Central Afr. Rep.	-0.117	0.208	112	36	-0.273	0.113	36	1
Chad	-0.044	0.176	164	13	.	.	.	1
Congo	0.115	0.045	340	46	0.118	0.068	46	1
Eq. Guinea	0.297	0.118	172	18	.	.	.	1
CEMAC average	0.115	0.088	1,401	249	0.072	0.188	218	5
High income	0.299	0.014	29,863	10,704	0.319	0.013	10,642	45
SSA excl. CEMAC	0.240	0.031	16,820	2,724	0.269	0.027	2,546	38

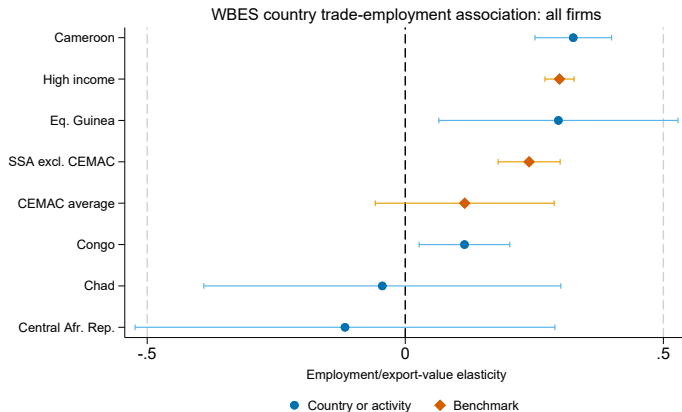
Slopes are weighted cross-sectional log employment/log export-value associations.

WBES Country Specification: All Firms

$$\ln(E_i) = \alpha_c + \beta_c \ln(X_i^+) + \gamma_c \text{Exporter}_i + \theta_c Z_i + \lambda_{cq(i)} + \varepsilon_i \quad (9)$$

- Estimated separately by country using WBES weights.
- Non-exporters stay in the sample with $\ln(X_i^+) = 0$.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{cq(i)}$ are country-specific ISIC Rev.4 section fixed effects.
- The plotted coefficient is β_c ; it is descriptive, not causal.

WBES Country Associations: All Firms



Weighted WBES cross-section. Points show log employment/log export-value associations; rows with N<30 suppressed.

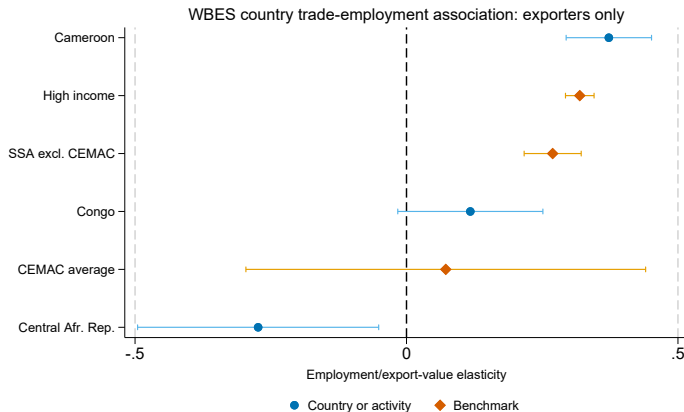
Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

WBES Country Specification: Exporters Only

$$\ln(E_i) = \alpha_c + \beta_c \ln(X_i) + \theta_c Z_i + \lambda_{cq(i)} + \varepsilon_i, \quad X_i > 0 \quad (10)$$

- Estimated separately by country among firms with positive export value.
- This is the intensive-margin association within exporters.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{cq(i)}$ are country-specific ISIC Rev.4 section fixed effects.
- Countries with fewer than 30 exporters are suppressed in the plot.

WBES Country Associations: Exporters Only



Weighted WBES cross-section. Points show log employment/log export-value associations; rows with N<30 suppressed.

Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Cameroon WBES Activity Estimates

	All-firm slope	SE	N	Exporters	Exporter-only slope	SE	N
Construction/utilities	-1.009	0.425	60	8	.	.	.
Manufacturing	0.482	0.066	159	51	0.545	0.069	51
Other services	0.428	0.079	52	9	.	.	.
Trade/hospitality/transport	0.287	0.050	342	68	0.363	0.047	68

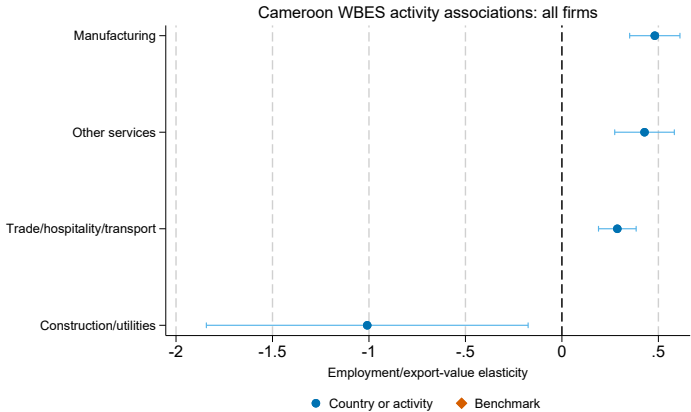
Activity groups use WBES-ISIC aggregates

Cameroon Activity Specification: All Firms

$$\ln(E_i) = \alpha_a + \beta_a \ln(X_i^+) + \gamma_a \text{Exporter}_i + \theta_a Z_i + \lambda_{aq(i)} + \varepsilon_i \quad (11)$$

- Estimated within Cameroon WBES activity groups.
- This is a diagnostic adaptation of the elasticity idea to cross-sectional survey data.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{aq(i)}$ are activity-specific ISIC Rev.4 section fixed effects where there is within-activity variation.
- Activity rows pass the all-firm sample threshold, but exporter-only rows may be suppressed.

Cameroon Activity Associations: All Firms



Weighted WBES cross-section. Points show log employment/log export-value associations; rows with N<30 suppressed.

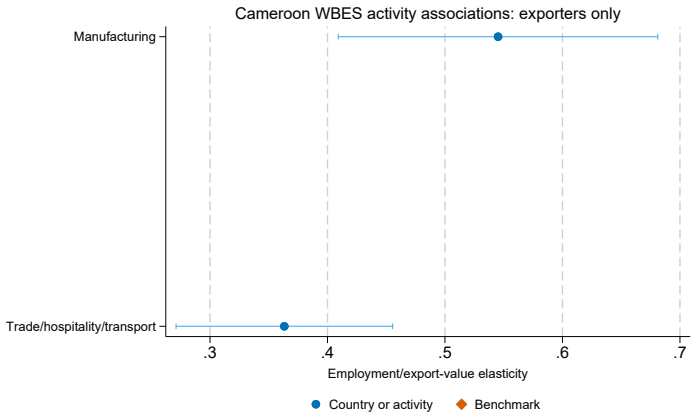
Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Cameroon Activity Specification: Exporters Only

$$\ln(E_i) = \alpha_a + \beta_a \ln(X_i) + \theta_a Z_i + \lambda_{aq(i)} + \varepsilon_i, \quad X_i > 0 \quad (12)$$

- Estimated among Cameroon exporters within WBES activity groups.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{aq(i)}$ are activity-specific ISIC Rev.4 section fixed effects where there is within-activity variation.
- Rows are shown only when at least 30 exporter observations are available.
- These results are best read as sample diagnostics for follow-up, not sector multipliers.

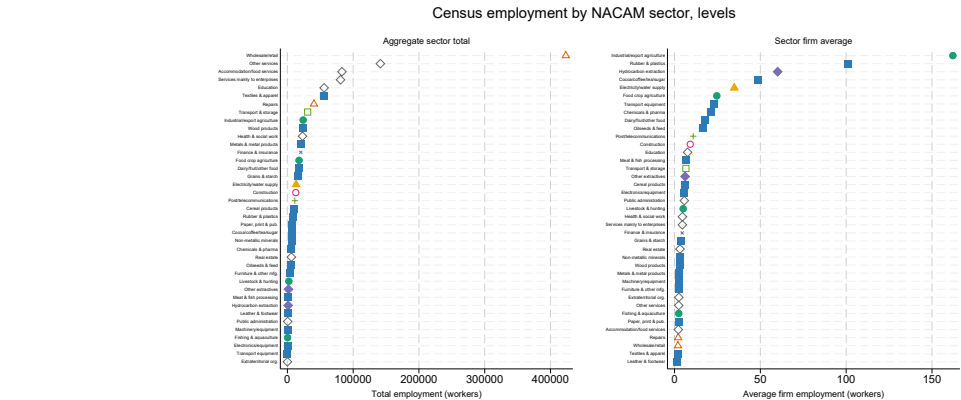
Cameroon Activity Associations: Exporters Only



Weighted WBES cross-section. Points show log employment/log export-value associations; rows with N<30 suppressed.

Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

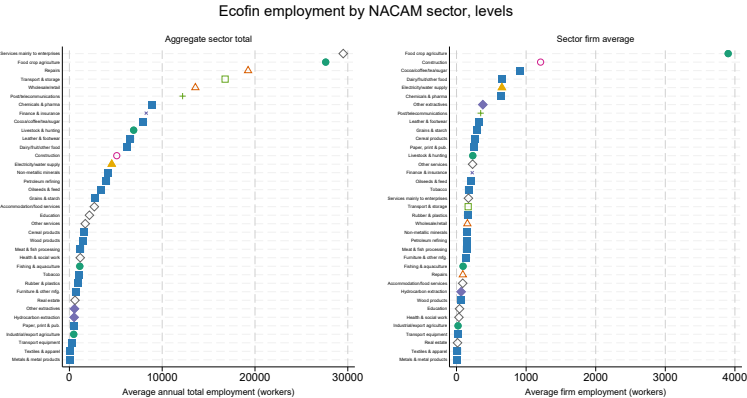
Census Employment by Harmonized NACAM Sector, Levels



Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

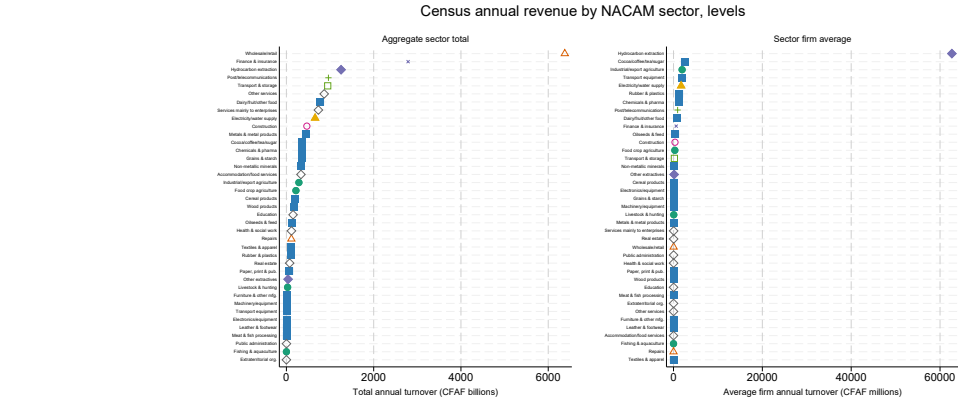
Ecofin Employment by Harmonized NACAM Sector, Levels



Back to log figure

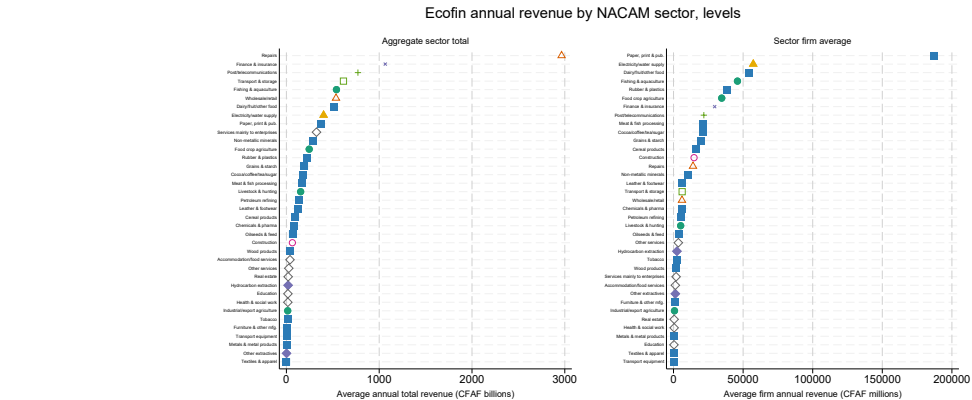
Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Revenue by Harmonized NACAM Sector, Levels



Back to log figure
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

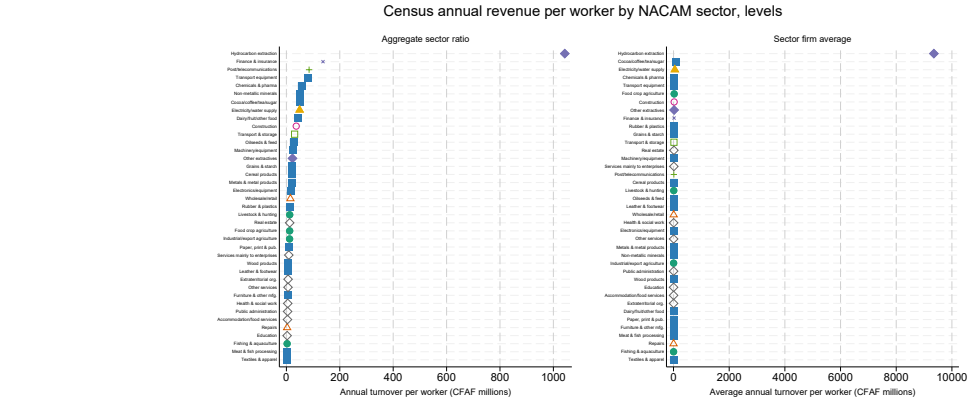
Ecofin Revenue by Harmonized NACAM Sector, Levels



Back to log figure

Source: Cameroon administrative Ecofin panel; calculations by authors.

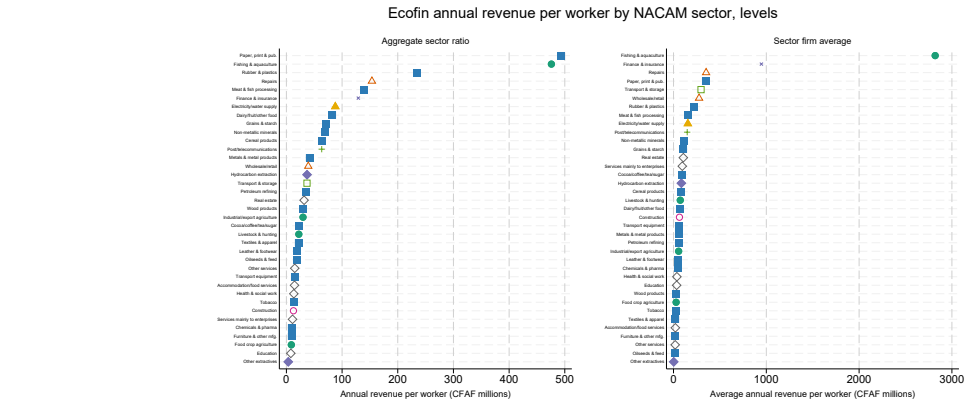
Census Revenue per Worker by Harmonized NACAM Sector, Levels



Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Revenue per Worker by Harmonized NACAM Sector, Levels



Back to log figure

Source: Cameroon administrative Ecofin panel; calculations by authors.